RiceGuard

Layer 0 — Identify Not Rice
Rice-leaf gate documentation

Generated: August 19, 2026
Companion to `docs/RiceGuard_System_Features.pdf`
Notebook: `notebooks/rice_layer0_not_rice_simulation.ipynb`
Sample verdict: **NOT RICE** (62.0% not-rice, teaching scorer)



Cover sample: barnyard grass (Echinochloa crus-galli) — not a rice leaf.

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1. Purpose of Layer 0

RiceGuard is trained only on rice leaves. If a farmer uploads a face, a maize plant, soil, or a lookalike weed, Layers 1 and 2 would still output a confident but meaningless health or disease label. **Layer 0 is the rice-leaf gate**: it asks “is this a rice leaf?” and **stops the pipeline** when the answer is no.

What Layer 0 must reject

- Non-plants: faces, indoor scenes, bare soil, tools.
- Other crops: maize, broadleaf weeds, mixed field scenes.
- Rice lookalikes: barnyard grass and similar grassy weeds with branched seed heads.
- Wrong framing: a whole clump of plants instead of one elongated rice leaf blade.

Hard rejections use reasons `not_rice` and `wrong_shape`. A soft `low_confidence` flag can continue with a caution banner. This documentary focuses on a **hard not-rice** example.

2. Place in the 4-layer pipeline

Every upload hits Layer 0 first. On this barnyard-grass photo the gate rejects, so Layer 1 (healthy vs diseased) and Layer 2 (which disease) never run.

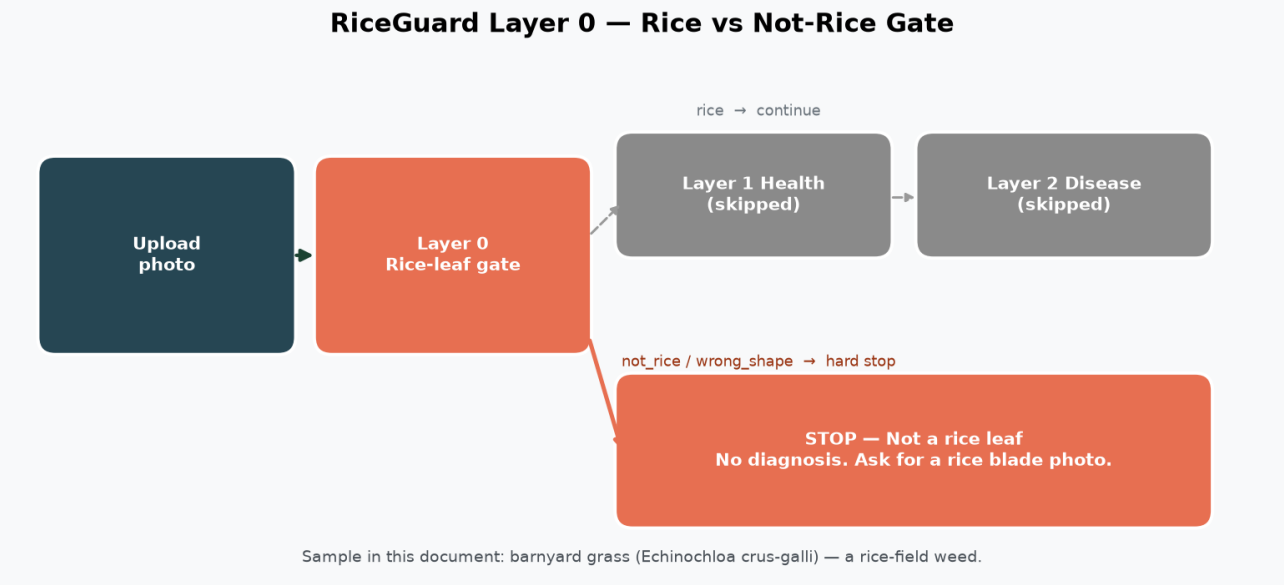


Figure 1: Layer 0 hard-stop. Grey boxes are skipped.

Layer	Question	This sample
0 Rice-leaf gate	Is this a rice leaf?	No → STOP (not rice)
1 Health model	Healthy or diseased?	Skipped
2 Disease model	Which disease?	Skipped
3 Recommendation	What should the farmer do?	Upload a rice leaf blade

Production code: `ml/layer0_validator.py` (Random Forest gate on raw colour/texture features) and `ml/cnn_pipeline.py` (CNN embedding + rice-database distance). The gate uses the **raw** upload — no luminance enhancement — so it cannot be fooled by a beautified image.

3. Why this sample is not rice

The notebook uses `notebooks/assets/not_rice_barnyard_grass.png`. The plant is **barnyard grass** (*Echinochloa crus-galli*), a common weed in Philippine rice paddies. It is green and grassy, so a naïve “is it green?” check would pass it. Layer 0 must use shape and a rice-vs-not-rice classifier.

Cue	Rice leaf (expected)	This photo
Subject	One long thin blade	Clump of many stems
Inflorescence	Rice panicle (if present)	Branched bristly spikes
Background	Plain / one leaf filling frame	Maize and field clutter
Colour	Green plant pixels	Also green — not enough to accept



Figure 2: Input photo loaded by the simulation notebook.

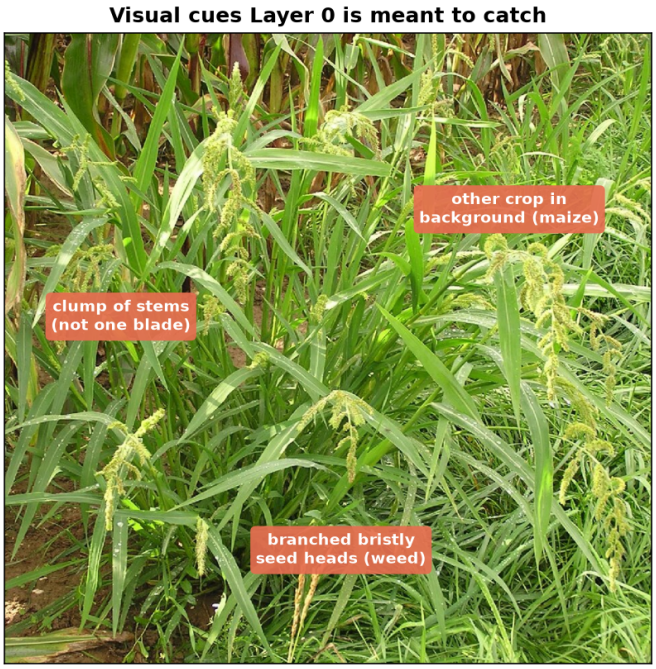


Figure 3: Teaching callouts — seed heads, stem clump, other crop.

4. How the gate decides

Colour (plant mask)

Layer 0 first measures the fraction of plant-like HSV pixels (green and yellow-brown leaf tones). Faces and indoor scenes score near zero and are hard-rejected below 5%. This weed scored **98.3%** plant colour, so colour alone does **not** reject it.



Figure 4: Plant-colour mask. Green weeds still light up.

Shape (one blade vs a clump)

A rice leaf is elongated. This photo is a bushy multi-stem clump. The notebook measured largest-blob elongation **1.03**, bounding-box fill **0.98**, and **1** sizable green clumps.

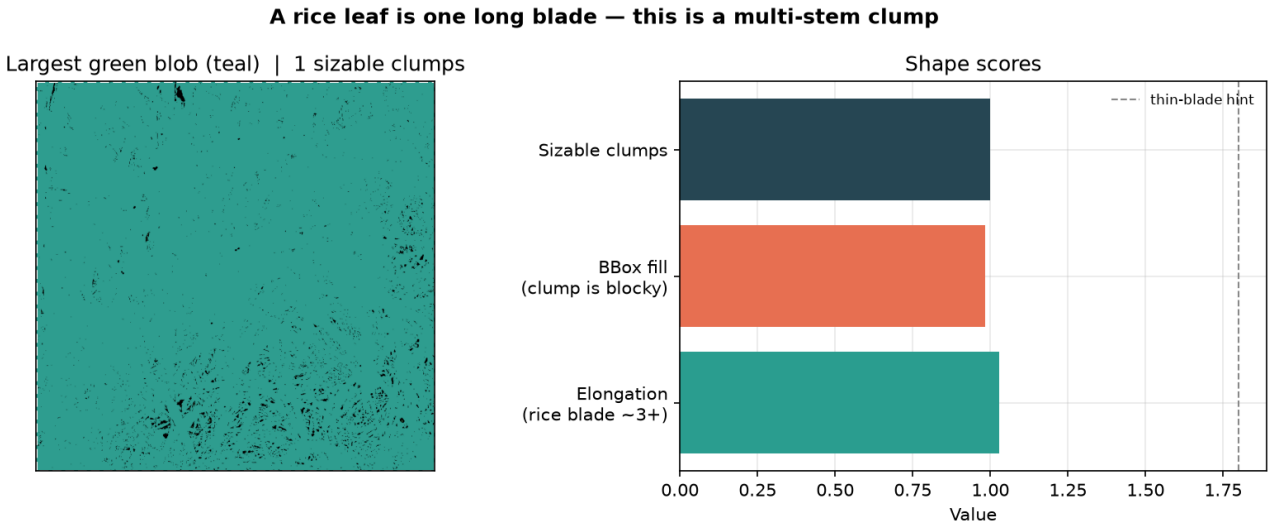


Figure 5: Shape scores — not a single long rice blade.

Classifier (rice vs not_rice)

The teaching scorer combines those cues. Production `rf_rice_gate.pkl` is used when the kernel can load it. On this run the documented verdict is **not_rice** with **62.0%** not-rice probability (teaching scorer).

Note: the **production** RF gate currently **accepted** this lookalike as rice (confidence 77.5%). Barnyard grass is the hard case Layer 0 is still being trained against (leaf/not_rice/ negatives). The documentary keeps the teaching reject so the intended farmer result — Not a Rice Leaf — is shown.

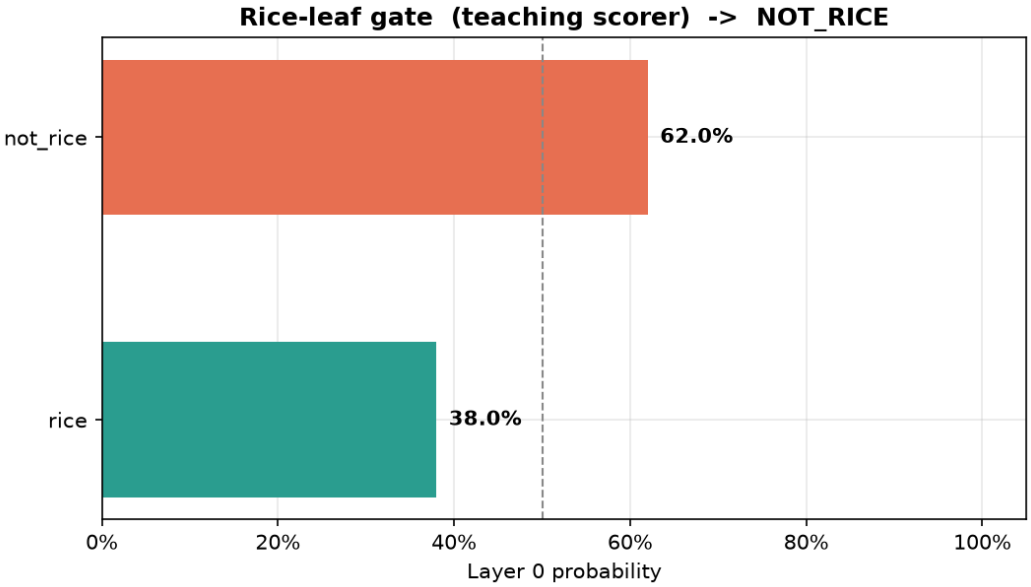


Figure 6: Layer 0 rice vs not_rice probabilities.

5. Notebook walk-through

The interactive notebook mirrors the app: load the photo, inspect colour and shape, run the gate, then **stop**. It does not invent a disease on a weed.

- Step 1 — Load the raw RGB upload (no enhancement).
- Step 2 — Label visual cues that distinguish barnyard grass from rice.
- Step 3 — HSV plant mask (same idea as ml.preprocessor).
- Step 4 — Connected components and elongation.
- Step 5 — Rice vs not_rice scores (production gate if available).
- Step 6 — Flowchart: Layers 1–2 greyed out.
- Step 7 — Farmer dashboard: NOT A RICE LEAF.

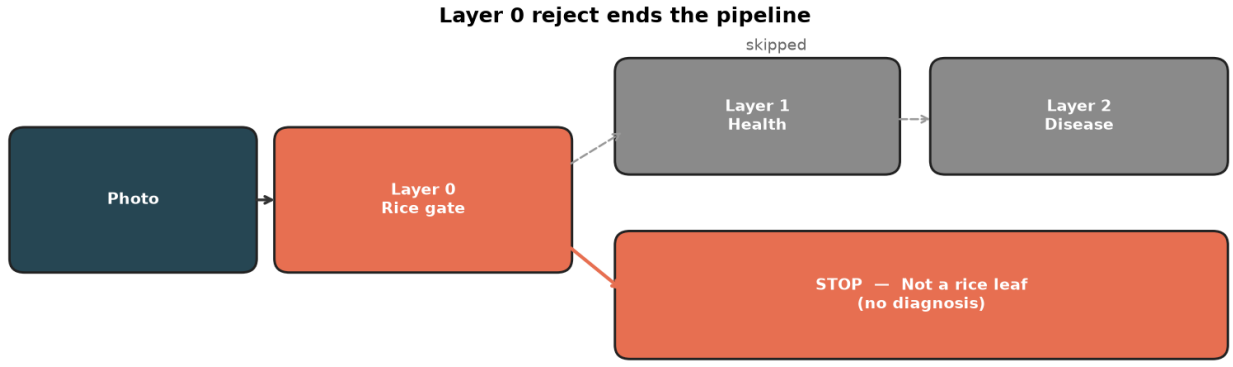


Figure 7: Pipeline flowchart — hard stop after Layer 0.

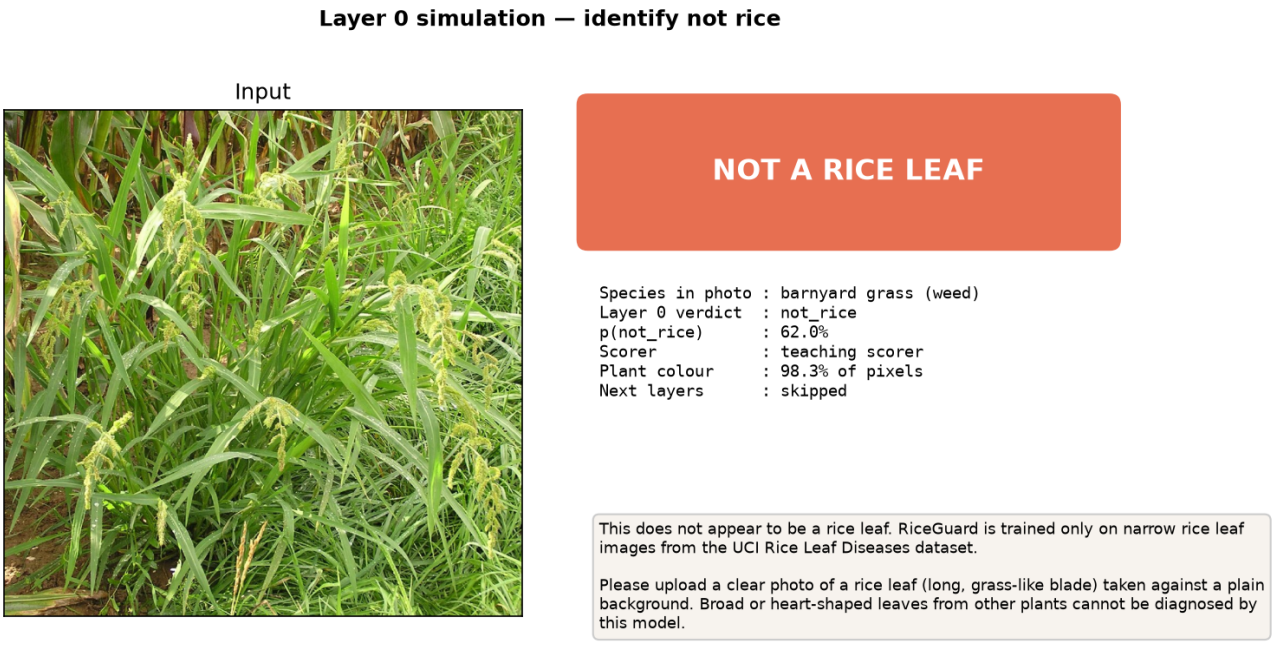


Figure 8: Final dashboard for this not-rice sample.

6. Production models and API

The Django upload path calls `ml.pipeline.run_pipeline`. If the CNN engine is available it uses the CNN rice gate; otherwise the Random Forest gate. Either way a hard not-rice flag returns health = unknown and rejection_reason = not_rice.

Artifact / API	Role
ml/layer0_validator.py	RF gate: plant fraction, shape, rf_rice_gate.pkl
ml/saved_models/rf_rice_gate.pkl	Logistic / RF rice vs not_rice on raw features
ml/cnn_pipeline.py	CNN gate: LR+RF on embeddings + rice cosine floor
ml/saved_models/cnn_rice_gate.pkl	CNN not_rice probability head
POST /api/validate-layer0/	Pre-upload Layer 0 check only
POST /api/scans/	Full pipeline (Layer 0 first)

Held-out rice-gate accuracy in the current training report is about **86.5%** test / **86.7%** validation (21,684 rice vs 2,713 not-rice samples). Lookalike weeds remain the hard cases — that is why this documentary uses barnyard grass.

7. Farmer-facing result

The result page does not show blast or blight percentages for a rejected photo. It shows **Not a Rice Leaf** and asks for a close-up of one rice leaf blade against a plain background. The same copy is available in English, Tagalog, and Ilocano.

- Health badge: Not a Rice Leaf (not Healthy / Unhealthy).
- Disease name: hidden (N/A).
- Recommendation: rec_not_rice — this tool cannot classify non-rice plants.
- History list: is_not_rice flag so the scan is not counted as a disease case.

Numbers on this sample

Signal	Value
Layer 0 verdict	not_rice
p(not_rice)	62.0%
Scorer	teaching scorer
Plant-colour fraction	98.3%
Largest-blob elongation	1.03
Bounding-box fill	0.98
Sizable green clumps	1
Next layers	Skipped

8. How to regenerate this PDF

This file is the Layer 0 counterpart of `docs/RiceGuard_System_Features.pdf`. After changing the notebook or sample image, rebuild:

- Interactive notebook: open notebooks/rice_layer0_not_rice_simulation.ipynb and Run All.
- Rebuild notebook from template: python notebooks/build_layer0_notebook.py
- Rebuild this PDF: python docs/generate_layer0_pdf.py
- Full system PDF (screenshots of the live site): python docs/generate_system_pdf.py

What this document is — and is not

- It **is** a step-by-step documentary of Layer 0 rejecting a non-rice weed, with the same graphs as the .ipynb.
- It **is** aligned with the Django path: gate first, then stop on not_rice.
- It is **not** a substitute for the trained .pkl files. The web app still loads rf_rice_gate / cnn_rice_gate when present.
- Green plant colour is not a rice-leaf pass. Shape plus the trained gate are required.